SKILL WALLET
A SmartBridge Product

SMARTBRIDGE
Let's Bridge the Gap

Problem-Solution Fit

Date	
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:
Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Smallholder farmers (primary) Agricultural extension officers Agri-tech policymakers Farming cooperatives	**6. CUSTOMER LIMITATIONS** EG. BUDGET, DEVICES **[CL]** Limited smartphone penetration in rural areas Low technical literacy Budget constraints for premium tools Irregular internet connectivity	**5. AVAILABLE SOLUTIONS** PROS & CONS **[AS]** Traditional knowledge — (Pro) Free (Con) Not data-driven Govt. extension services — (Pro) Trusted (Con) Slow, limited reach Manual soil labs — (Pro) Accurate (Con) Expensive, slow turnaround
2. PROBLEMS / PAINS + ITS FREQUENCY **[PR]** Low crop yield from poor crop selection (every season) Soil degradation — monocropping (every 2-3 yrs) Wasted input costs on	**9. PROBLEM ROOT / CAUSE [RC]** No data-driven decision-making tools accessible to smallholders Lack of centralized soil & climate analysis	**7. BEHAVIOR** + ITS INTENSITY **[BE]** Rely on inherited practices & peer advice (HIGH — affects livelihood) Visit agri officers only when problems escalate (MEDIUM)

unsuitable seeds/fertilizers (every cycle) Climate uncertainty affecting harvest (annually)	Inability to match crop requirements with local environmental conditions	Trial-and-error planting across seasons (HIGH intensity)
3. TRIGGERS TO ACT [TR] Repeated crop failure or low yield Declining soil fertility observed over seasons Price fluctuations making current crop unprofitable Government advisories on changing weather patterns	**10. YOUR SOLUTION [SL]** OptiCrop — ML-powered crop recommendation engine. • Analyzes 7 parameters: N, P, K, temperature, humidity, pH, rainfall • Random Forest classifier (99.5% accuracy) predicts best crop • Provides suitability analysis with soil health & environmental insights • Intuitive web interface — accessible from any device • Designed for farmers, extension officers & policymakers	**8. CHANNELS OF BEHAVIOR [CH]** ONLINE • Smartphone / basic web browsers • Agri-information portals • Social media farmer groups
4. EMOTIONS BEFORE / AFTER **[EM]** BEFORE: Anxious about seasonal yield — frustrated by unpredictable outcomes — stressed by financial risk of wrong crop choice. AFTER: Confident in data-backed decisions — empowered with actionable insights — relieved knowing they optimized for local conditions.		OFFLINE • Local agricultural extension offices • Farmer cooperative meetings • Field demonstration days